Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Inertial frame

Meaning in my words How would you explain inertial frame to a friend?	**How can you tell?** A puck glides straight at steady speed. Would a camera moving straight at constant speed also see steady motion?
Example or sketch Which bus ride could serve as an inertial frame for watching a puck: cruising straight or turning?	**Non-example and why** A bus brakes and a loose puck seems to surge forward inside it. Does that prove a forward force acted on the puck?

Net force

Meaning in my words How would you explain net force to a friend?	**How can you tell?** Two people push one cart in opposite directions. Why do the arrow directions matter?
Example or sketch One person pushes 6 N right and another 4 N left. What single arrow replaces both?	**Non-example and why** A student adds those pushes to get 10 N right. Which push did they count the wrong way?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Inertia

Meaning in my words What does inertia mean? Say it the way you'd explain it to a friend.	**How can you tell?** An empty cart and a loaded cart both start at rest. We push each with the same net force for one second. Which gains less speed? Why?
Example or sketch A cart is already rolling right. The forces on it balance. Where would you draw it one second later? And one second after that?	**Non-example and why** Someone points to the hand pushing the cart and says, "That push is inertia." What would you tell them?

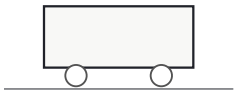
Equilibrium

Meaning in my words How would you explain equilibrium to a friend?	**How can you tell?** A cart rolls straight at steady speed. What must the horizontal forces add up to?
Example or sketch Sketch the horizontal forces on a cart with a 3 N pull right and 3 N friction left.	**Non-example and why** A tossed ball is briefly still at the top. Is it in equilibrium there?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a moving cart with balanced force arrows in orange and its separate velocity arrow in purple.

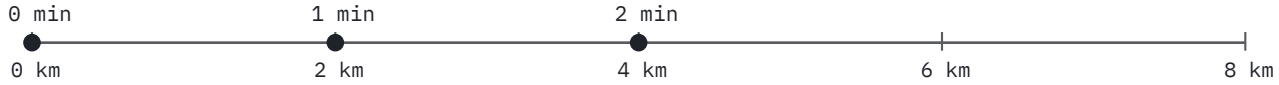


My marks mean

Where should we point the telescope?

SECTION E / FINAL MEETING Name _____

The telescope spots XJ once a minute. At minute 3, the camera can take one picture. Your team gets to choose where it points.



Camera windows A +4 km B +6 km C +8 km

Choose a job together. One of you marks the prediction. The other makes the call.

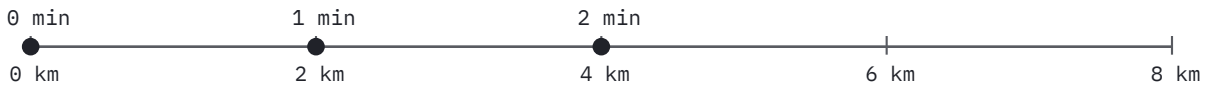
☐ **Tracking engineer**

Circle the window you'd aim for. Tell the camera operator where to point and which two sightings helped you decide.

☐ **Live science broadcaster**

Tell viewers where to look. Someone asks, "If XJ has no engine, why does it keep moving?" Use your cart drawing to answer.

Mark your prediction here.



WHEN THE NEXT SIGHTING COMES IN

Mark the new sighting. How far did XJ move in the last minute? Switch speakers and tell your crew or your viewers what changed.

Why could a straight-line forecast work for a minute even while gravity pulls on XJ?